

Predicting Credit Default

Kernel Approximation and Stable Neural Training

Leonardo Pelorosso

Person code: 10779110

Numerical Analysis for Machine Learning (NAML)

Abstract. Regularized logistic regression, a Nyström kernel SVM and a JAX MLP are compared on five stratified 70/15/15 splits. Feature spectra, Hessian curvature and landmark eigenvalue decay connect conditioning to regularization and optimization. SVM/MLP mean average precision is $0.3991 \pm 0.0083/0.4003 \pm 0.0036$; the MLP has the lowest mean Brier score (0.0490 ± 0.0003).

1 Data and Experimental Protocol

Give Me Some Credit contains 150,000 rows and ten borrower attributes. The binary target indicates serious delinquency within two years. Each run uses 105,000 training, 22,500 validation and 22,500 test rows. For seeds $s \in \{42, 52, 62, 72, 82\}$, the stratified split steps use s and $s + 1$. Seed 42 supplies the figures; its training prevalence is 6.684%. Average precision (AP), the step-weighted precision–recall summary, is used instead of accuracy alone; it is not trapezoidal PR-AUC.

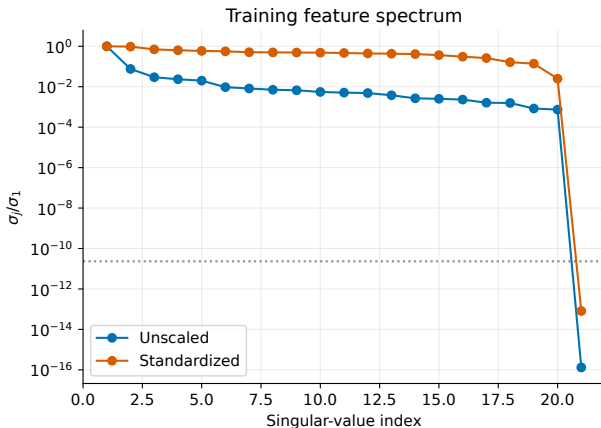
The shared preprocessing produces 21 features using training medians, missingness flags, fixed clipping, logarithms and three interaction ratios. Delinquency codes 96/98 are flagged and replaced by zero; the income/debt transformation is a proxy, not recovered monetary debt. Medians and standardization are refitted on each run’s training rows only.

Each split repeats its own bounded search. Validation supports selection, MLP stopping, SVM sigmoid calibration and thresholds; this reuse can make its scores optimistic. All choices precede that split’s test predictions; no Train+Validation refit or ensemble is used. Overlapping test sets make the five-seed sample standard deviation descriptive, not a confidence interval.

Feature conditioning

For a full-column-rank feature matrix X ,

$$\kappa_2(X^T X) = \left(\frac{\sigma_{\max}(X)}{\sigma_{\min}(X)} \right)^2. \quad (1)$$



The total delinquency count is a sum of three other columns: both unscaled and standardized matrices have numerical rank 20, so $\kappa_2(X^T X) = \infty$. SVD identifies this dependency without relying on the smallest eigenvalues of a rounded Gram matrix. Restricting the ratio to numerically nonzero singular values gives 1.82×10^6 before standardization and 1.59×10^3 after it (Figure 1). Scaling improves this restricted conditioning; it does not remove the dependency.

2 Logistic Regression

With $z = w^T x + b$ and $p = \sigma(z)$, minimize mean binary cross-entropy plus $\lambda \|w\|_2^2/2$, leaving the intercept unpenalized. The stable loss is

$$\ell(z, y) = \text{logaddexp}(0, (1 - 2y)z). \quad (2)$$

SciPy `minimize` uses L-BFGS-B and the analytic gradient. Six penalties $\lambda = 10^k$, $k = -6, \dots, -1$, define the bounded grid. Validation AP selects $\lambda = 10^{-1}$ for seed 42: 11 accepted iterations, training/validation BCE 0.19439/0.19688 and validation AP 0.35750.

With $\tilde{X} = [X \ \mathbf{1}]$, $D = \text{diag}(p_i(1 - p_i))$ and $R = \text{diag}(1, \dots, 1, 0)$, the training Hessian is

$$H_\lambda = \tilde{X}^T D \tilde{X} / N + \lambda R. \quad (3)$$

L2 lifts flat coefficient directions without penalizing the intercept. Across the grid, $\kappa_2(H_\lambda)$ falls from 4.72×10^5 to 13.68 and iterations from 50 to 11 (Figure 2). Curvature is measured at each fitted solution; it alone does not determine an L-BFGS convergence rate.

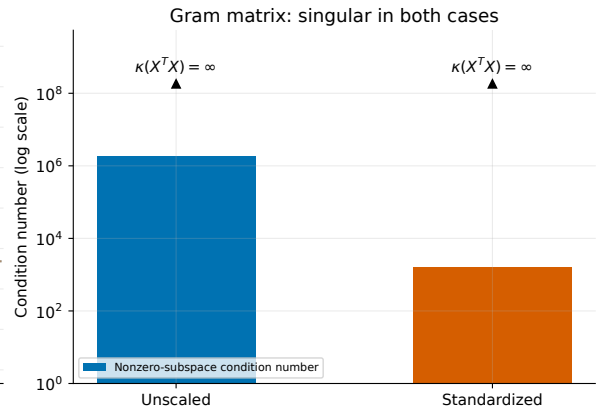


Figure 1: Seed-42 training feature spectrum. Both Gram matrices are singular; finite bars report only nonzero-subspace condition numbers. Dotted spectrum lines mark the SVD rank tolerance.

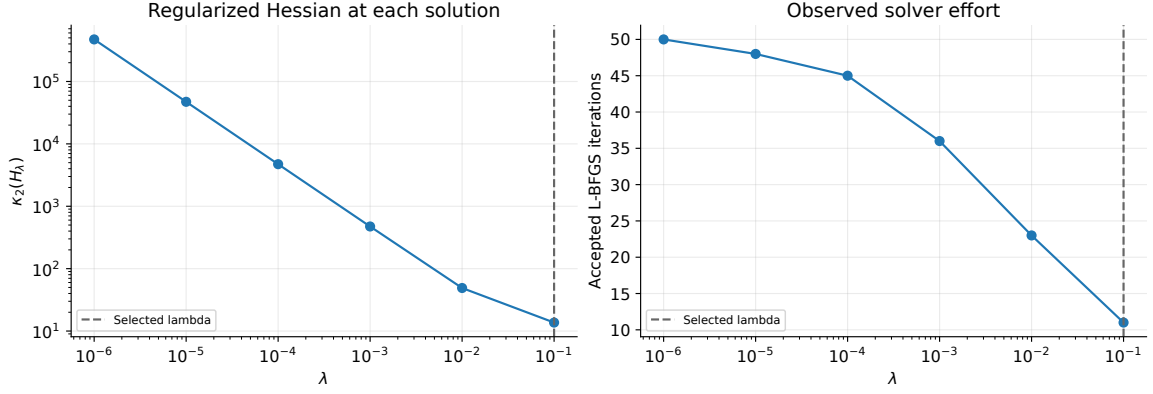


Figure 2: Regularized logistic Hessian conditioning and observed L-BFGS effort. The dashed line marks the validation-selected penalty.

3 Nyström Kernel SVM

An RBF kernel $k(x, z) = \exp(-\gamma\|x - z\|_2^2)$ represents nonlinear similarity. For m training landmarks L , set $W = K(L, L) = Q\Lambda Q^T$ and $C_X = K(X, L)$. The approximate features are [1]

$$\Phi_\epsilon(X) = C_X Q(\Lambda + \epsilon I)^{-1/2}. \quad (4)$$

Thus $\tilde{K} = \Phi_\epsilon \Phi_\epsilon^T$ approximates the full kernel without storing its $N \times N$ matrix. The spectral shift ϵ stabilizes small eigenvalues; it differs from classifier regularization λ .

For $t_i = 2y_i - 1$ and $f_i = w^T \phi_i + b$, the linear SVM on these features minimizes

$$J = \frac{1}{N} \sum_i \max(0, 1 - t_i f_i)^2 + \frac{\lambda}{2} \|w\|_2^2. \quad (5)$$

Squared hinge loss permits an analytic-gradient L-BFGS solver. The joint grid has 48 candidates per split. Landmark counts are $m = 64, 128, 256, 512$; bandwidth scales are $s \in \{1/16, 1/4, 1\}$. Penalties are $\lambda = 10^k$ for $k = -5, \dots, -2$. Here $\gamma = s / \text{median}(\|x_i - x_j\|^2)$ uses training pairs only. Landmark sets are nested prefixes of one training-row permutation per seed.

For seed 42, validation AP selects $m = 512$, $s = 1/4$, $\gamma = 0.01065$ and $\lambda = 10^{-5}$. Training/validation squared hinge loss is 0.19461/0.19787 after 615 accepted iterations (Figure 3). A nonnegative-slope sigmoid fitted on validation maps scores to probabilities

[2]; validation AP is 0.38759. All solver termination statistics are saved to CSV.

Kernel approximation and conditioning

The numerical sweep holds the selected bandwidth and classifier penalty fixed; it does not replace joint model selection. From $m = 64$ to 512, relative Frobenius error decreases from 0.008108 to 0.003796 and validation AP increases from 0.37923 to 0.38759. Lower reconstruction error need not improve prediction for a different kernel or landmark sample.

The landmark spectrum (Figure 4) exposes small eigenvalues. These four matrices retain full numerical rank at the implemented tolerance, but become severely ill-conditioned. For a positive-definite shifted matrix,

$$\kappa_2(W + \epsilon I) = \frac{\lambda_{\max}(W) + \epsilon}{\lambda_{\min}(W) + \epsilon}. \quad (6)$$

For the selected map, $\epsilon = 3.80 \times 10^{-6}$ reduces the condition number from 2.23×10^{11} to the enforced target 10^8 . An eigensolver can decompose a singular matrix; the difficulty is amplification when tiny eigenvalues are inverted in Eq. (4). Jitter controls this amplification while introducing approximation bias. No dense inverse is formed.

Factorization costs $O(m^3)$, dense projection $O(Nm^2)$ and feature storage $O(Nm)$, plus $O(Nmd)$ kernel evaluation for $d = 21$ features.

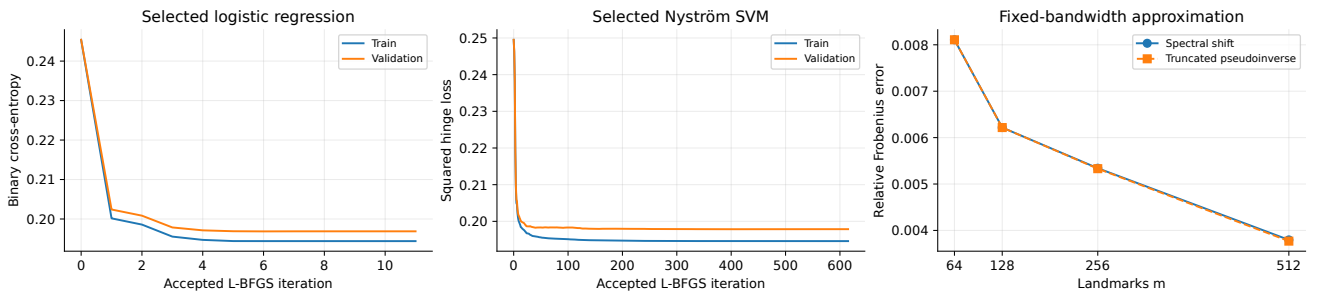


Figure 3: Seed-42 selected logistic/SVM training and validation losses, excluding L2 penalties. Right: relative kernel reconstruction error on the same 1,200 validation rows, at fixed selected bandwidth and nested landmark sets. Search-run learning curves are omitted.

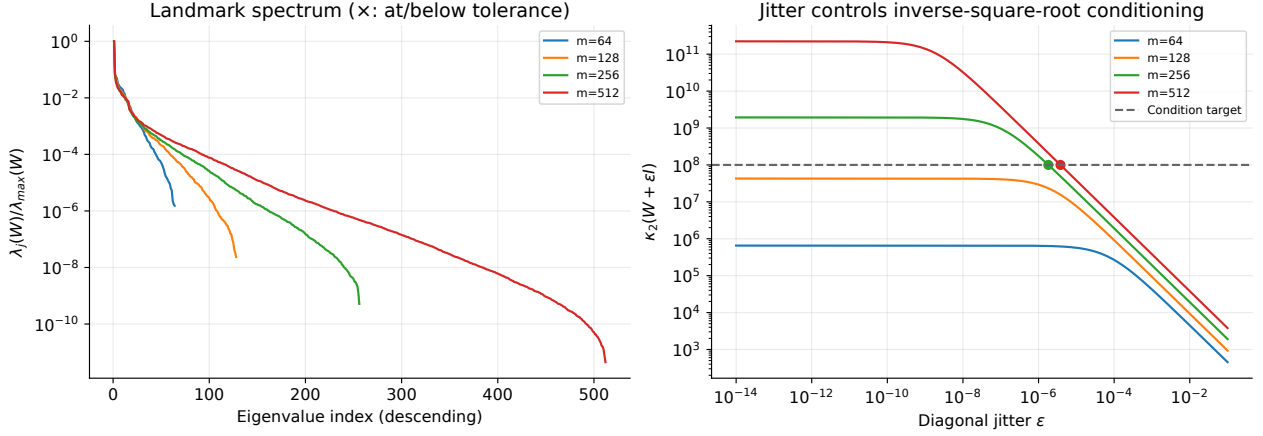


Figure 4: Landmark eigenvalue decay across m and the condition number of $W + \epsilon I$ versus positive jitter. Dots mark nonzero shifts selected by the numerical safeguard; the dashed line is its 10^8 target.

4 JAX MLP: Shallow versus Deep

One hidden layer of width 54 and two hidden layers (32, 16) contain 1,243 and 1,249 parameters. Both use ReLU, Glorot initialization, Adam [3] and batches of 256. The grid crosses these architectures with learning rates $\{3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}\}$ and L2 strengths $\{10^{-6}, 10^{-5}, 10^{-4}\}$: 18 candidates per split. This compares tuned predictors, not an isolated effect of depth.

The objective is the unweighted mean of Eq. (2) plus $\lambda \sum_l \|W_l\|_F^2/2$, with unpenalized biases. JAX float64 remains enabled. Loss evaluation from logits avoids overflow and logarithms of rounded probabilities. L2 enters Adam’s gradient without another $1/N$ scaling. Early stopping uses validation BCE, tolerance 10^{-5} , patience 8 and at most 60 epochs; validation AP selects the configuration, with BCE breaking ties.

For seed 42, the selected shallow MLP uses learning rate 10^{-3} and $\lambda = 10^{-4}$. It runs 26 epochs and restores epoch 18; validation BCE is 0.17913 and AP is 0.38557

(Figure 5). Only the selected run is plotted. Relative Adam updates diagnose step sizes, not convergence to a global minimum.

4.1 Automatic and numerical differentiation

Reverse-mode AD computes the scalar loss gradient through the chain rule. It has floating-point round-off but no finite-difference truncation error. For a unit parameter direction v , the central difference

$$d_h = \frac{\mathcal{L}(\theta + hv) - \mathcal{L}(\theta - hv)}{2h} \quad (7)$$

approximates $\nabla \mathcal{L}(\theta)^T v$. Smooth-region error combines $O(h^2)$ truncation and approximately $O(u/h)$ round-off. The retained synthetic check reaches relative error 6.32×10^{-11} at $h = 3.16 \times 10^{-6}$; smaller steps lose accuracy. ReLU mask changes are marked separately. Extreme-logit and singular-landmark checks also pass; they validate numerical components, not global neural-network convergence.

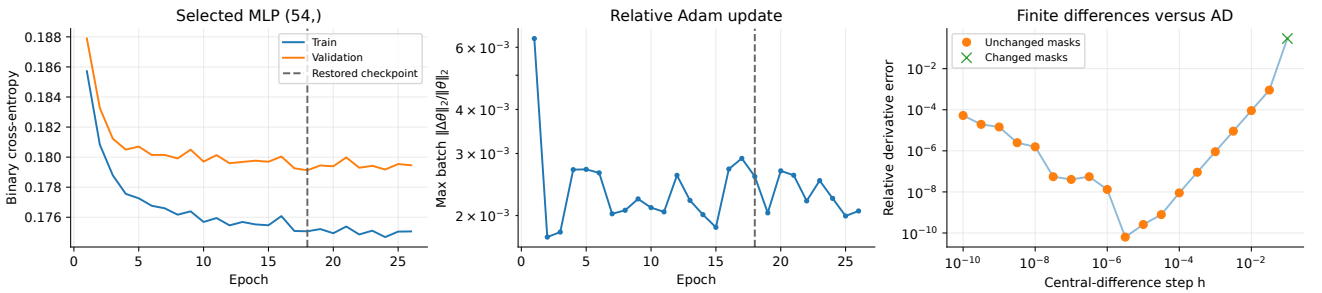


Figure 5: Selected seed-42 MLP loss and maximum relative Adam update per epoch; dashed lines identify the restored checkpoint. Right: central differences versus the AD directional derivative on the unchanged small synthetic network.

5 Final Test Comparison

The same 6/48/18 candidate grids are searched separately in all five runs, varying data splits, MLP initialization/shuffling and landmark sampling together. Logistic regression is deterministic on a fixed split. Each run fits training-only preprocessing, then freezes its selected models, calibration and validation-cost thresholds before evaluating 22,500 test cases, including 1,504 defaults. Selected settings can vary across seeds; seed 42 is reused rather than refitted.

Table 1: Out-of-sample mean $\pm$ sample standard deviation across five seeds. Higher AP/ROC-AUC and lower Brier/cost are better. Cost is per 1,000 cases, using each run’s own validation-selected threshold. These overlapping holdouts are not independent replicates.

Model	AP	ROC-AUC	Brier	Cost 5:1	Cost 10:1
Logistic regression	0.3774 $\pm$ 0.0047	0.8529 $\pm$ 0.0042	0.0520 $\pm$ 0.0002	218.92 $\pm$ 2.19	349.41 $\pm$ 8.74
Nyström SVM	0.3991 $\pm$ 0.0083	0.8635 $\pm$ 0.0054	0.0503 $\pm$ 0.0003	214.38 $\pm$ 1.75	333.70 $\pm$ 8.40
JAX MLP	0.4003 $\pm$ 0.0036	0.8637 $\pm$ 0.0054	0.0490 $\pm$ 0.0003	213.67 $\pm$ 2.11	335.17 $\pm$ 6.47

Both nonlinear methods have higher mean AP than logistic regression. The MLP has slightly higher mean AP/ROC-AUC and lower mean Brier and 5:1 cost; the SVM has lower mean 10:1 cost. Their small differences do not establish statistical superiority. The reported variability combines split, selection and training randomness; it does not isolate their contributions. Log loss and raw per-seed metrics remain available in the saved CSV records.

6 Threshold Selection and Misclassification Costs

Predict default when $p \geq t$. False negatives (FN) miss defaults; false positives (FP) flag non-defaults. With zero cost for correct classifications,

$$C(t) = c_{FN}FN(t) + c_{FP}FP(t). \quad (8)$$

For calibrated probabilities, default/non-default decisions have expected costs $c_{FP}(1 - p)$ and $c_{FN}p$. Constant costs therefore imply

$$t^* = \frac{c_{FP}}{c_{FP} + c_{FN}}. \quad (9)$$

The illustrative ratios 5:1 and 10:1 give $t^* = 1/6$ and

1/11; they are not measured bank losses. Empirical thresholds minimize validation cost and are frozen before testing (Figure 6). F1 thresholds do not encode these costs. Miscalibration and sampling effects can separate empirical and theoretical rules. For $c_{FP} = 1$ and $c_{FN} = r$, empirical expected cost per 1,000 cases is

$$\hat{C}_{1000} = \frac{1000}{n}(rFN + FP). \quad (10)$$

Table 1 aggregates this quantity across seeds; Table 2 retains the seed-42 error counts as a concrete decision example.

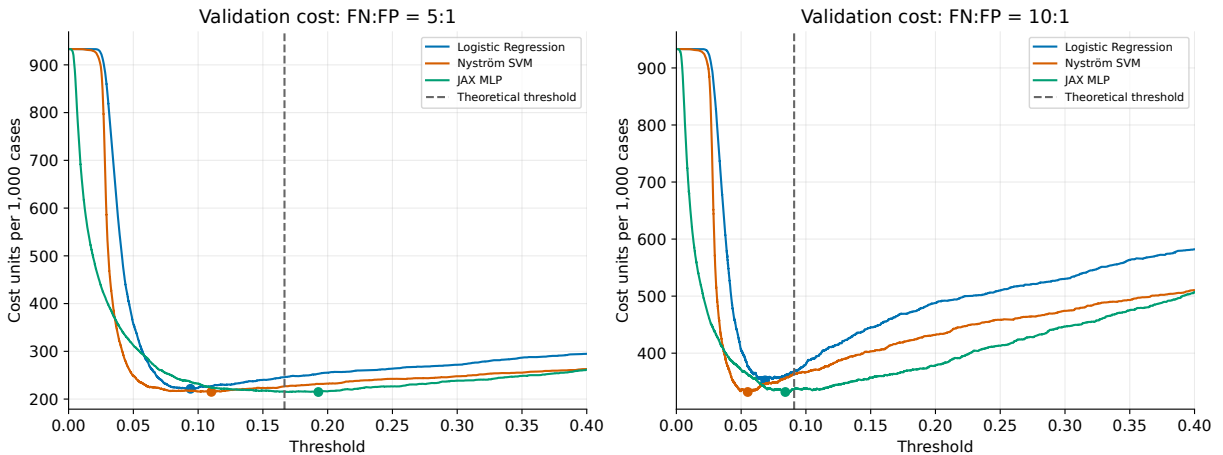


Figure 6: Seed-42 validation cost per 1,000 cases at two assumed FN:FP ratios. Dots mark frozen empirical minima; dashed lines show theoretical thresholds. The threshold axis is restricted to $[0, 0.4]$ for readability; the search covers all unique validation probabilities and both endpoint policies.

Table 2: Representative seed-42 test outcomes of validation-selected cost policies. Costs are per 1,000 cases; precision and recall are percentages. Displayed thresholds are rounded; predictions use their full saved precision.

FN:FP	Model	Threshold	FP	FN	Precision	Recall	Cost/1,000
5:1	Logistic regression	0.0941	1,650	639	34.4	57.5	215.33
	Nyström SVM	0.1102	1,328	690	38.0	54.1	212.36
	JAX MLP	0.1928	1,238	717	38.9	52.3	214.36
10:1	Logistic regression	0.0685	3,019	455	25.8	69.7	336.40
	Nyström SVM	0.0551	3,107	414	26.0	72.5	322.09
	JAX MLP	0.0841	3,450	395	24.3	73.7	328.89

Interpreting the decision trade-off

For the seed-42 MLP at 10:1, threshold 0.5 gives 17.6% recall and cost 557.38 per 1,000 cases; the F1 threshold 0.1928 gives 52.3% recall and cost 373.69. The cost-selected threshold 0.0841 raises recall to 73.7% and lowers cost to 328.89, with more false positives. This is 50.8% below the always-non-default cost of 668.44.

The theoretical MLP threshold 1/11 costs 326.76 on this test split, slightly below the validation-selected rule; no test-based retuning follows. The SVM has the lowest empirical-policy cost at both ratios in Table 2, whereas five-seed mean cost favors the MLP at 5:1 and SVM at 10:1 (Table 1). AP alone cannot determine the best decision policy.

Conclusions and limitations

Standardization improves nonzero-subspace conditioning but leaves an exact feature dependency; L2 supplies missing coefficient curvature. The spectral shift controls Nyström inverse-square-root conditioning, and logits-based BCE supports stable JAX training. AD and solver checks establish numerical consistency, not predictive optimality. Tuning both architectures supports no general claim about a depth advantage.

Five repeated holdouts replace the single-run com-

parison, but jointly vary splits, selection, initialization and landmarks. Validation reuse can cause optimism, and overlapping test sets preclude interpreting the sample SD as a confidence interval. Clipping, debt proxies and costs remain assumptions. Previously inspected test results make this an exploratory numerical study; independent calibration assessment would strengthen its conclusions.

Reproducibility. The notebook preserves float64 and records split identifiers, package versions, a dataset checksum and frozen policies. Full grids, convergence logs and solver termination statistics go to `artifacts_focused/`; raw results are in `five_seed_study.csv`. All 360 grid candidates and 378 fit/termination records completed with no solver failures or execution warnings. Only selected-run learning curves are displayed.

References

- [1] C. K. I. Williams and M. Seeger. Using the Nyström method to speed up kernel machines. *Advances in NIPS 13*, 2001.
- [2] J. C. Platt. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. *Advances in Large Margin Classifiers*, 1999.
- [3] D. P. Kingma and J. Ba. Adam: A method for stochastic optimization. *ICLR*, 2015.